Experimental Evaluation of RAG Component Design Choices over an Information Retrieval Textbook Corpus

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as the dominant architecture for knowledge-intensive NLP tasks, yet the impact of individual design choices on system quality remains under-studied for domain-specific corpora. This paper presents a systematic experimental evaluation of four key RAG components — chunking strategy, embedding model, retrieval method, and LLM backend — applied to the Manning et al. Introduction to Information Retrieval textbook. Using a hand-crafted evaluation set of 20 questions and metrics including Precision@K, Mean Average Precision, and LLM-as-judge faithfulness scoring, we find that: (1) paragraph-based chunking best preserves semantic coherence for structured academic text; (2) local MiniLM embeddings are competitive with OpenAI API embeddings at zero cost; (3) hybrid dense+sparse retrieval achieves the highest MAP (0.883); and (4) GPT-4o-mini provides near-optimal faithfulness (4.60/5) at 40x lower cost than Claude Sonnet. These findings offer practical guidance for building cost-effective RAG systems over technical document corpora.

1 INTRODUCTION

Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide range of natural language tasks. However, they suffer from two fundamental limitations when applied to domain-specific question answering: (1) a fixed knowledge cutoff that excludes documents unavailable at training time, and (2) a tendency to hallucinate — generating plausible but factually incorrect responses when the required knowledge is absent.

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020), addresses both limitations by retrieving relevant document passages at query time and grounding the LLM's response in the retrieved context. RAG has since become the standard architecture for enterprise knowledge bases, document Q&A systems, and academic research assistants.

Despite its widespread adoption, RAG is not a single algorithm but a family of architectures with many interchangeable components. Practitioners must choose: how to split documents into chunks, which embedding model to use, whether to retrieve by dense vector similarity, sparse keyword matching, or a combination, and which LLM to use for generation. Each choice carries tradeoffs in quality, latency, and cost. Yet comparative studies across all four dimensions on a single corpus remain scarce.

This paper systematically varies each component while holding the others constant, measuring the effect on retrieval and generation quality. Our corpus is the Manning, Raghavan, and Schutze Introduction to Information Retrieval textbook — a 569-page academic reference that is thematically aligned with our course and provides a challenging, structured evaluation domain.

Our main contributions are: (1) a reproducible experimental framework for RAG component evaluation; (2) empirical results across all four design dimensions; and (3) a practical recommendation for cost-effective RAG over technical academic corpora. All code is available at: *github.com/Yakira3012/ir-final-assignment-course3*.

2 RELATED WORK

The foundational RAG architecture was proposed by Lewis et al. (2020), who demonstrated that combining a dense retriever (DPR) with a seq2seq generator outperformed closed-book models on knowledge-intensive tasks. Subsequent work explored retrieval improvements: Karpukhin et al. (2020) showed that dense passage retrieval surpasses BM25 on open-domain QA, while Ma et al. (2022) demonstrated that hybrid dense-sparse retrieval consistently outperforms either method alone.

On chunking, recent work by Shi et al. (2023) showed that lost-in-the-middle effects — where information at the edges of long contexts is better utilized than information in the middle — argue for smaller, more focused chunks. Robertson and Zaragoza (2009) provide the theoretical foundation for BM25, the sparse retrieval method used in this work.

Unlike prior work, which typically evaluates a single component or uses general web corpora, this paper evaluates all four components together on a specialized academic textbook domain.

3 SYSTEM DESCRIPTION

3.1 Corpus and Preprocessing

Our corpus is the Manning et al. Introduction to Information Retrieval textbook (569 pages, approximately 200,000 words). The PDF was parsed using pdfplumber, followed by cleaning steps including page number removal, hyphenation correction, and whitespace normalization. The cleaned text contains approximately 1.1 million characters (~291,000 tokens).

3.2 Experiment A: Chunking Strategies

We evaluated three chunking strategies. Fixed-size chunking splits text every 256 tokens with a 50-token overlap, producing consistent chunk sizes but ignoring linguistic boundaries. Sentence-based chunking groups complete sentences until reaching a 256-token target, preserving grammatical units. Paragraph-based chunking splits on double newlines, using the textbook's own structural boundaries, with short paragraphs merged (minimum 50 tokens) and long ones split (maximum 512 tokens).

Table 1: Chunking strategy statistics.

Strategy	# Chunks	Mean (tok)	Median (tok)	Max (tok)
Fixed-size	498	726	704	1,544
Sentence-based	1,199	243	245	3,429
Paragraph-based	1,058	275	244	552

Paragraph-based chunking produces fewer, larger chunks aligned with textbook structure.

3.3 Experiment B: Embedding Models

We compared two embedding models applied to the paragraph-based chunks. The all-MiniLM-L6-v2 model (Wang et al., 2020) is a lightweight transformer that runs locally, producing 384-dimensional L2-normalized vectors in approximately 82 seconds on CPU at zero cost. OpenAI's text-embedding-3-small produces 1,536-dimensional vectors via API call, offering higher capacity at a cost of approximately $0.007 for our corpus.

Table 2: Embedding model comparison.

Model	Dims	Time (s)	Cost (USD)	Source
all-MiniLM-L6-v2	384	81.6	**$0.000**	Local
text-embedding-3-small	1536	118.1	$0.007	OpenAI API

MiniLM offers competitive retrieval quality at zero cost; OpenAI provides higher-dimensional representations.

3.4 Experiment C: Retrieval Methods

We implemented three retrieval methods over the paragraph chunks embedded with MiniLM. Dense retrieval uses a FAISS IndexFlatIP index with inner product similarity (equivalent to cosine similarity on normalized vectors). Sparse retrieval uses BM25Okapi from rank-bm25 (k1=1.5, b=0.75). Hybrid retrieval combines both via a normalized weighted sum with alpha=0.5, where each score distribution is first normalized to [0, 1]:

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score_hybrid = 0.5 * score_dense_norm + 0.5 * score_sparse_norm
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3.5 Experiment D: LLM Generation Backends

Retrieved chunks are assembled into a prompt with a strict instruction to answer only from the provided context. We evaluated three LLM backends: llama3 via Ollama running locally on a GPU machine (zero cost, ~102s latency), claude-sonnet-4-20250514 via Anthropic API ($0.160 total, ~6.2s latency), and gpt-4o-mini via OpenAI API ($0.004 total, ~1.5s latency).

4 EVALUATION

4.1 Evaluation Set

We constructed a hand-crafted evaluation set of 20 questions drawn from the IR textbook, comprising 10 factual questions (e.g., formula definitions), 6 conceptual questions (e.g., comparing algorithms), and 4 applied questions (e.g., tracing an algorithm on an example). Each question includes a reference answer and the source chapter.

4.2 Retrieval Metrics

We report Precision@3 (P@3), Precision@5 (P@5), and Mean Average Precision (MAP) over the 20 questions. A retrieved chunk is considered relevant if it originates from the source chapter of the corresponding question. MAP averages the area under the precision-recall curve across all queries and is our primary ranking metric.

4.3 Generation Metrics

Generation quality is evaluated using an LLM-as-judge approach: Claude Haiku scores each generated answer on a 1-5 faithfulness scale, judging whether the answer is supported by the retrieved context. We also report average latency per query and total cost across all 20 questions per backend.

5 RESULTS

5.1 Experiment A: Chunking

Paragraph-based chunking produced the strongest downstream retrieval performance. Its chunks align with the textbook's own conceptual divisions, resulting in semantically coherent units. Fixed-size chunking occasionally split mid-concept, reducing chunk quality. Sentence-based chunks were grammatically clean but sometimes too granular for technical content. All subsequent experiments use paragraph-based chunks.

5.2 Experiment B: Embeddings

MiniLM and OpenAI embeddings produced comparable downstream retrieval P@3 scores (difference < 0.02). Given MiniLM's zero cost and local execution, it was selected as the embedding model for subsequent experiments. This finding suggests that for technical academic text with consistent vocabulary, a lightweight local model can match a high-dimensional API model.

5.3 Experiment C: Retrieval

Table 3: Retrieval method evaluation (20 questions).

Retriever	P@3	P@5	MAP
Dense (FAISS)	0.800	0.820	0.846
Sparse (BM25)	0.700	0.730	0.839
Hybrid	**0.800**	**0.770**	**0.883**

Hybrid retrieval achieves highest MAP; Dense and Hybrid tie on P@3.

Dense retrieval outperforms sparse on P@5 (0.820 vs 0.730), confirming that semantic matching better handles paraphrased questions. BM25 is competitive on factual questions containing exact technical terms. Hybrid retrieval achieves the best MAP (0.883), validating the complementary nature of dense and sparse signals. The P@5 result for

Hybrid (0.770) being lower than Dense suggests some rank dilution at deeper cutoffs when combining scores from different distributions.

5.4 Experiment D: LLM Generation

Table 4: LLM backend evaluation (20 questions, top-3 hybrid retrieval).

Backend	Faithfulness	Avg Latency	Total Cost	Source
Claude Sonnet 4	4.65 / 5	6.2s	$0.160	API
GPT-4o-mini	**4.60 / 5**	1.5s	$0.004	API
Ollama / llama3	4.20 / 5	102s	$0.000	Local

Claude Sonnet achieves the highest faithfulness; GPT-4o-mini is near-equivalent at 40x lower cost.

All three backends achieve high faithfulness scores (>4.2/5), confirming that RAG effectively grounds generation when context quality is high. Claude Sonnet marginally outperforms GPT-4o-mini (4.65 vs 4.60) but at 40x higher cost ($0.160 vs $0.004) and 4x higher latency. Ollama's lower faithfulness (4.20) and 102-second latency make it impractical for interactive use despite zero cost.

5.5 Best Pipeline

The winning configuration is: paragraph chunking + MiniLM embeddings + hybrid retrieval + GPT-4o-mini generation. This pipeline achieves MAP=0.883 and faithfulness=4.60 at an estimated cost of $0.004 per 20 queries — making it suitable for production deployment. Claude Sonnet is recommended when maximum faithfulness is required and cost is not a constraint.

6 DISCUSSION

Our results support three practical recommendations for RAG system design over academic corpora. First, use structure-aware chunking: paragraph boundaries in textbooks encode the author's own semantic groupings and consistently outperform arbitrary fixed-size splits. Second, hybrid retrieval is robust: it combines the semantic coverage of dense retrieval with the precision of BM25 for exact technical terms, achieving the best MAP without any additional fine-tuning. Third, retrieval quality is the bottleneck: all three LLMs achieve similar faithfulness when given high-quality retrieved context, suggesting that investment in better retrieval yields higher returns than upgrading the LLM.

A notable finding is that MiniLM matches OpenAI embeddings in downstream retrieval quality for this domain. We attribute this to the consistent, formal vocabulary of the IR textbook — unlike conversational or multilingual text where larger models show clearer advantages.

7 LIMITATIONS

Several limitations apply to this work. The evaluation set of 20 questions, while carefully crafted across question types, is small. A larger annotated test set with explicit relevance judgments at the chunk level would enable more reliable P@K estimation. The faithfulness metric relies on Claude Haiku as an automated judge, which may introduce systematic bias toward Claude-generated answers. Human evaluation of a subset of answers would strengthen this metric. Finally, results are specific to the IR textbook domain; generalization to other corpora (e.g., legal documents, code repositories) requires additional study.

8 ETHICS STATEMENT

The corpus used in this work — Manning et al. Introduction to Information Retrieval — is freely available for academic use at the URL provided in the course syllabus. No personal data was collected or used. API usage (OpenAI, Anthropic) was conducted under standard terms of service. The automated faithfulness judge (Claude Haiku) is used only for research evaluation and not for any consequential decision-making. All code is open-source under the MIT license.

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